

Lectures Review: Generalization and Classification

Yingming PU

LINs Lab, Westlake University

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Outline

① Validation

② Bias–Variance

③ The Sigmoid

④ Cross-Entropy

⑤ Summary

Train / val / test: why we split data

We partition the dataset S into three disjoint roles:

- **Train:** Fits parameters of candidate models $\hat{f}_1, \dots, \hat{f}_K$.

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The Central Question:

What is a validation score actually worth — and how close is it to true risk?

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How far is the **computable sample average** from the **true population mean**?

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- **Fresh points:** Validation points sampled i.i.d. from $\mathcal{D} \implies Z_i$ are **i.i.d.**
- **Bounded range:** Loss bounded in $[a, b] \implies Z_i \in [a, b]$.

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The Statistical Question:

How far can an average of m i.i.d. bounded numbers $\bar{Z}$ wander from its mean μ ?

Lemma: Markov's inequality I

We begin with the most primitive tool for bounding tail probabilities:

Markov's inequality. For any random variable $X \geq 0$ and any $t > 0$:

$$\Pr(X \geq t) \leq \frac{\mathbb{E}[X]}{t}.$$

Proof. Notice that $X \geq t \cdot \mathbf{1}\{X \geq t\}$ (check both $X \geq t$ and $X < t$). Taking expectations on both sides yields $\mathbb{E}[X] \geq t \cdot \Pr(X \geq t)$. □

Apply directly to deviation from the mean:

$$\text{Set } X = |\bar{Z} - \mu| : \quad \Pr(|\bar{Z} - \mu| \geq \epsilon) \leq \frac{\mathbb{E}|\bar{Z} - \mu|}{\epsilon}.$$

Lemma: Markov's inequality II

Diagnosis: Markov inequality is too crude...

- **Decay rate $1/\epsilon$ is painfully slow:** to reduce tolerance ϵ by $10\times$, the bound gets $10\times$ looser, with no exponential sample concentration.
- **Key problem structures are completely wasted:** In our setting, Z_i are **independent** and **bounded** in $[a, b]$. Yet raw Markov uses *neither*!

Next move: to unlock independence.

Lemma: the exponential trick

Why the exponential transform? We need $\phi(\sum_i A_i) = \prod_i \phi(A_i)$ so independence can split the expectation — only $\phi(u) = e^{tu}$ does this ($t > 0$). With $\bar{Z} = \frac{1}{m} \sum_i Z_i$, $\mu = \mathbb{E}[Z_i]$:

$$\Pr(\bar{Z} - \mu \geq \epsilon) = \Pr(e^{tm(\bar{Z}-\mu)} \geq e^{tm\epsilon})$$

$$\leq e^{-tm\epsilon} \mathbb{E}[e^{tm(\bar{Z}-\mu)}]$$

$$= e^{-tm\epsilon} \mathbb{E}\left[\exp\left(t\sum_{i=1}^m (Z_i - \mu)\right)\right]$$

$$= e^{-tm\epsilon} \mathbb{E}\left[\prod_{i=1}^m e^{t(Z_i - \mu)}\right]$$

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$e^{(\cdot)}$ is strictly increasing

Markov bound (tail factor emerges)

definition of $\bar{Z}$

$$e^{\sum a_i} = \prod e^{a_i}$$

independence splits the expectation

The tail is now factored into decay $e^{-tm\epsilon}$ and **one 1-D factor** to the m -th power: $\mathbb{E}[e^{t(Z-\mu)}]^m$.

Lemma: Hoeffding's lemma I

Hoeffding's lemma. If $Z \in [a, b]$ and $\mathbb{E}[Z] = 0$, then $\mathbb{E}[e^{tZ}] \leq e^{t^2(b-a)^2/8}$.

Proof. Write $p := -a/(b-a)$ and $u := t(b-a)$ as we go.

$$e^{tZ} \leq \frac{b-Z}{b-a} e^{ta} + \frac{Z-a}{b-a} e^{tb}$$

$$\mathbb{E}[e^{tZ}] \leq \frac{b}{b-a} e^{ta} + \left(-\frac{a}{b-a}\right) e^{tb}$$

$$= (1-p) e^{-pu} + p e^{(1-p)u}$$

$$=: e^{\phi(u)}, \quad \phi(u) = -pu + \ln(1-p+pe^u)$$

$e^{t(\cdot)}$ is convex: secant line above curve

take $\mathbb{E}$ (note $\mathbb{E}[Z] = 0$)

substitute $p = \frac{-a}{b-a}$, $u = t(b-a)$

factor out e^{-pu} into the exponent

Now bound ϕ itself. Differentiate twice:

Lemma: Hoeffding's lemma II

$$\phi(0) = 0, \quad \phi'(0) = 0$$

direct substitution

$$\phi''(u) = \frac{p(1-p)e^u}{(1-p+pe^u)^2}$$

two rounds of the quotient rule

$$= \frac{pe^u}{1-p+pe^u} \cdot \frac{1-p}{1-p+pe^u} \leq \frac{1}{4}$$

$$xy \leq \frac{(x+y)^2}{4}, \quad x+y=1$$

$$\phi(u) = \phi(0) + \phi'(0)u + \frac{1}{2}\phi''(\xi)u^2 \leq \frac{u^2}{8}$$

Taylor expansion with remainder

Substituting back $u = t(b-a)$ gives the statement. □

This is the **only** place $[a, b]$ enters the whole derivation:

bounded support $[a, b]$ guarantees the quadratic upper bound on $\phi(t)$.

Without bounded support, exponential sub-Gaussian concentration fails.

Conclusion (assembly): plug in, then choose the bend t

Plug step 3 into step 2. Each of the m factors is bounded by $e^{t^2(b-a)^2/8}$:

$$\begin{aligned} \Pr(\bar{Z} - \mu \geq \epsilon) &\leq e^{-tm\epsilon} \prod_{i=1}^m e^{t^2(b-a)^2/8} && \text{step 3, factor by factor} \\ &= \exp\left(-tm\epsilon + \frac{mt^2(b-a)^2}{8}\right) && \text{merge the } m \text{ factors} \end{aligned}$$

Choose the best bend t . The exponent is a quadratic in t ; set the derivative to zero:

$$\begin{aligned} \frac{d}{dt} \left(-tm\epsilon + \frac{mt^2(b-a)^2}{8} \right) &= -m\epsilon + \frac{mt(b-a)^2}{4} = 0 \quad \implies \quad t^* = \frac{4\epsilon}{(b-a)^2} \\ -t^*m\epsilon + \frac{m(t^*)^2(b-a)^2}{8} &= -\frac{2m\epsilon^2}{(b-a)^2} \end{aligned}$$

Conclusion (assembly): Hoeffding's inequality

The lower tail is identical (replace Z_i by $-Z_i$), so the two tails add a factor 2:

Hoeffding's inequality. For $Z_i \in [a, b]$ i.i.d.,

$$\Pr(|L_{S_{\text{val}}}(f) - L_{\mathcal{D}}(f)| \geq \epsilon) \leq 2 \exp\left(-\frac{2m\epsilon^2}{(b-a)^2}\right).$$

The $O(1/\sqrt{m})$ confidence interval originates from inverting the quadratic exponent in ϵ :
sub-Gaussian decay implies root-sample scaling.

Inverting Hoeffding: sample complexity and confidence

Setting tail probability to $\delta \in (0, 1)$ yields the confidence interval:

$$|L_{S_{\text{val}}}(f) - L_{\mathcal{D}}(f)| \leq (b - a) \sqrt{\frac{\ln(2/\delta)}{2m}} \quad \text{with probability } \geq 1 - \delta.$$

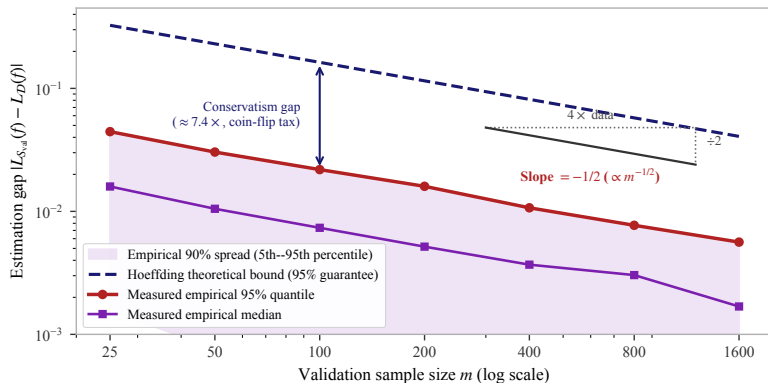
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A validation score is statistically bounded by the confidence interval: $L_{S_{\text{val}}}(f) \pm (b - a) \sqrt{\frac{\ln(2/\delta)}{2m}}$.

Living case: the measured gap follows the Hoeffding shape



The empirical gap scales strictly as $O(1/\sqrt{m})$ with sample size;
the gap between bound and empirical measurement reflects the conservatism of distribution-free guarantees.

Comparing K candidates: uniform convergence via union bound

For K hypotheses chosen independently of S_{val} , Boole's inequality yields:

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Uniform Hoeffding Bound.

$$\Pr \left(\max_{k \leq K} |L_{S_{\text{val}}}(f_k) - L_{\mathcal{D}}(f_k)| \geq \epsilon \right) \leq 2K \exp \left(- \frac{2m\epsilon^2}{(b-a)^2} \right) =: \delta.$$

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Inverting for the uniform deviation margin: $\epsilon(m, K, \delta) = (b-a) \sqrt{\frac{\ln(2/\delta) + \ln K}{2m}}$.

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Uniform concentration across K fixed models incurs only a mild $\sqrt{\ln K}$ penalty.

Prerequisite: Every candidate model f_k must be determined **independently of** S_{val} .

Selection Bias: optimization breaks sample independence

Selecting the empirical minimizer $\hat{k} = \arg \min_k L_{S_{\text{val}}}(f_k)$ violates sample independence: the selected index $\hat{k}$ is itself a random variable coupled with S_{val} .

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Jensen's Inequality on the Minimum Operator. The minimum function is concave:

$$\mathbb{E}_{S_{\text{val}}} \left[\min_{k \leq K} L_{S_{\text{val}}}(f_k) \right] \leq \min_{k \leq K} \mathbb{E}_{S_{\text{val}}} \left[L_{S_{\text{val}}}(f_k) \right] = \min_{k \leq K} L_{\mathcal{D}}(f_k).$$

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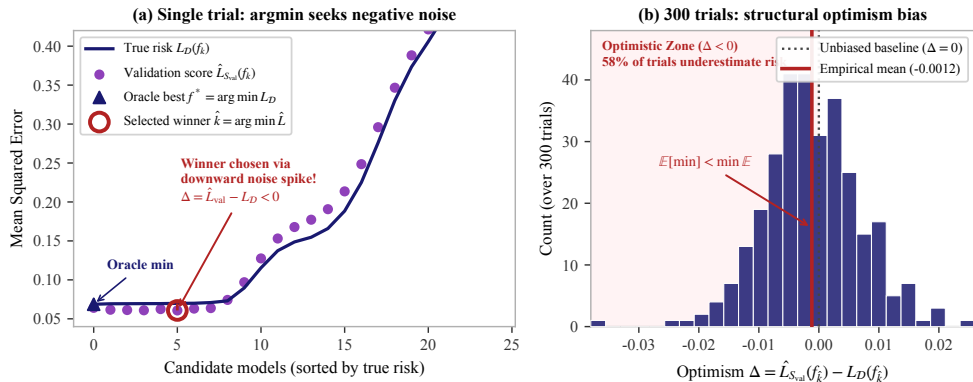
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A validation set evaluated to select models loses unbiasedness for risk estimation.
An independent **Test Set** is mathematically necessary for unbiased performance evaluation.

Living case: empirical optimism under model selection, measured



Model selection identifies high-performing candidates, but induces downward bias in empirical validation scores. Employ validation data for model selection, and evaluate exclusively on independent test data.

Recap: structural principles of statistical validation

- **1. Non-asymptotic confidence bounds:** Pointwise deviation is bounded by $L_{S_{\text{val}}}(f) \pm (b - a) \sqrt{\frac{\ln(2/\delta)}{2m}}$ via sub-Gaussian concentration.

Recap: structural principles of statistical validation

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- **3. Empirical selection bias:** Minimizing empirical loss couples the hypothesis with data, inducing structural optimism ($\mathbb{E}[\min] \leq \min \mathbb{E}$).

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- **4. Decoupled evaluation protocol:** Training set for parameter optimization $\longrightarrow$ Validation set for model selection $\longrightarrow$ Test set for unbiased risk reporting.

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- 1 Validation
- 2 Bias–Variance**
- 3 The Sigmoid
- 4 Cross-Entropy
- 5 Summary

The thought experiment: sampling variation of estimators

Fix a query point $x_0 \in \mathcal{X}$. Ground truth generates responses with noise:

$$y_0 = f(x_0) + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2.$$

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The Statistical Question:

How does the expected squared error decompose across random draws of S ?

The expected estimator: center of the model ensemble

To understand how $f_S(x_0)$ fluctuates, define its ensemble center:

Expected Estimator (Ensemble Mean):

$$\bar{f}(x_0) := \mathbb{E}_S[f_S(x_0)].$$

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$\bar{f}(x_0)$ is deterministic with respect to S — the hypothetical average predictor if nature generated infinitely many independent training datasets.

Lemma (Step 1): Isolating irreducible noise I

Express pointwise prediction error as label noise plus estimation error:

$$y_0 - f_S(x_0) = \varepsilon + (f(x_0) - f_S(x_0)).$$

Lemma (Step 1): Isolating irreducible noise II

- Query observation noise ε is generated independently of training set S :

$$\varepsilon \perp S \implies \mathbb{E}_{S, \varepsilon} [\varepsilon g(S)] = \mathbb{E}[\varepsilon] \cdot \mathbb{E}_S [g(S)].$$

- Since noise is zero-mean ($\mathbb{E}[\varepsilon] = 0$), the cross-term vanishes identically:

$$\mathbb{E}_{S, \varepsilon} \left[\varepsilon (f(x_0) - f_S(x_0)) \right] = 0.$$

$$\mathbb{E}_{S, \varepsilon} \left[(y_0 - f_S(x_0))^2 \right] = \sigma^2 + \mathbb{E}_S \left[(f(x_0) - f_S(x_0))^2 \right]$$

σ^2 is the **irreducible noise floor**:

it represents the fundamental information-theoretic lower bound on predictive risk.

Lemma (Step 2): Decomposing estimator error I

Center estimator error around the expected predictor $\bar{f}(x_0)$:

$$\underbrace{f(x_0) - \bar{f}(x_0)}_{\text{structural bias (deterministic)}} + \underbrace{(\bar{f}(x_0) - f_S(x_0))}_{\text{sample fluctuation (zero-mean)}} .$$

Lemma (Step 2): Decomposing estimator error II

- The bias $f(x_0) - \bar{f}(x_0)$ is deterministic with respect to S .
- Fluctuations around the mean vanish identically by definition of $\bar{f}$:

$$\mathbb{E}_S[\bar{f}(x_0) - f_S(x_0)] = \bar{f}(x_0) - \mathbb{E}_S[f_S(x_0)] = 0.$$

- **Approximation Bias:** $\text{bias}^2(x_0) := (f(x_0) - \bar{f}(x_0))^2$.
- **Estimation Variance:** $\text{Var}(x_0) := \mathbb{E}_S[(f_S(x_0) - \bar{f}(x_0))^2]$.

Mean-centering isolates the deterministic approximation gap (bias^2)
from the stochastic sensitivity to finite sample draws (Var).

Theorem: The Bias–Variance Decomposition

Combining Step 1 and Step 2 yields the pointwise decomposition:

Bias–Variance Decomposition Theorem. Under squared error loss:

$$\mathbb{E}_{S, \varepsilon} \left[(y_0 - f_S(x_0))^2 \right] = \underbrace{\sigma^2}_{\text{irreducible noise}} + \underbrace{\text{bias}^2(x_0)}_{\text{approximation error}} + \underbrace{\text{Var}(x_0)}_{\text{estimation variance}} .$$

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Integrating over the query input distribution $X \sim \mathcal{D}_X$:

$$\mathbb{E}_S [L_{\mathcal{D}}(f_S)] = \sigma^2 + \mathbb{E}_X [\text{bias}^2(X)] + \mathbb{E}_X [\text{Var}(X)] .$$

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Combining Step 1 and Step 2 yields the pointwise decomposition:

Bias–Variance Decomposition Theorem. Under squared error loss:

$$\mathbb{E}_{S, \varepsilon} \left[(y_0 - f_S(x_0))^2 \right] = \underbrace{\sigma^2}_{\text{irreducible noise}} + \underbrace{\text{bias}^2(x_0)}_{\text{approximation error}} + \underbrace{\text{Var}(x_0)}_{\text{estimation variance}} .$$

Integrating over the query input distribution $X \sim \mathcal{D}_X$:

$$\mathbb{E}_S [L_{\mathcal{D}}(f_S)] = \sigma^2 + \mathbb{E}_X [\text{bias}^2(X)] + \mathbb{E}_X [\text{Var}(X)] .$$

Generalization error under squared loss decomposes into three additive statistical components:
aleatoric noise, structural bias, and sample variance.

Geometric insight: Pythagorean orthogonality in Hilbert space

In the Hilbert space $L_2(\mathcal{D}_{X,Y})$ equipped with inner product $\langle u, v \rangle = \mathbb{E}[u \cdot v]$:

$$\|y_0 - f_S(x_0)\|^2 = \|\varepsilon\|^2 + \|f(x_0) - \bar{f}(x_0)\|^2 + \|\bar{f}(x_0) - f_S(x_0)\|^2.$$

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The three components are mutually orthogonal vectors:

- $(y_0 - f(x_0)) \perp (f(x_0) - \bar{f}(x_0))$: aleatoric noise is orthogonal to every deterministic function of x .

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The vanishing cross-terms are not an algebraic accident: they represent the exact **Pythagorean theorem** in function space.

The governing trade-off: model capacity vs sample size

Two independent factors control the magnitude of bias and variance:

- **1. Hypothesis Complexity $\mathcal{H}$ (Model capacity):**
 - Increasing capacity $\longrightarrow$ $\text{bias}^2 \downarrow$ (richer expressive space).
 - Increasing capacity $\longrightarrow$ $\text{Var} \uparrow$ (more parameters fitted to sample noise).

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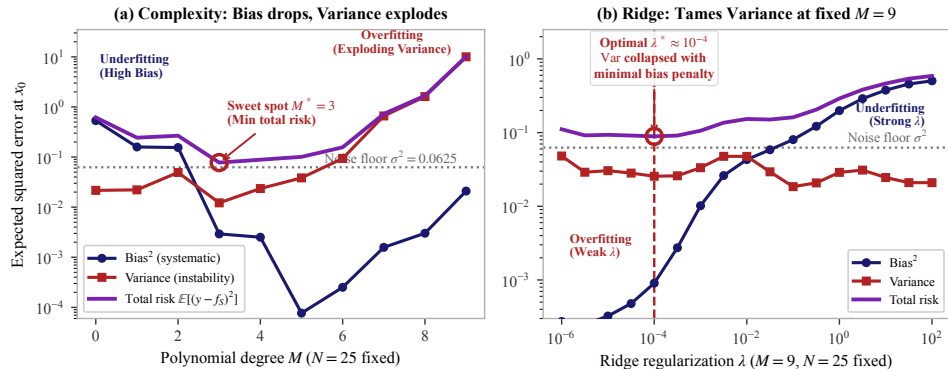
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Capacity shifts the balance between bias and variance for fixed N ;
expanding sample size N unilaterally shrinks variance without shifting bias.

Living case: empirical verification of the trade-off



Capacity sweeps trace the classical trade-off; L_2 regularization ($\lambda^* \approx 10^{-4}$) suppresses variance by two orders of magnitude while preserving approximation capacity.

(FYI) Modern perspective: classical trade-off vs double descent I

1. Underparameterized regime ($P < N$):

- Gram matrix $\mathbf{X}^\top \mathbf{X}$ is well-conditioned when $P \ll N$.

(FYI) Modern perspective: classical trade-off vs double descent II

2. Overparameterized regime ($P > N$) and Double Descent:

- For $P > N$, infinitely many parameter vectors interpolate the sample: $\mathbf{X}\mathbf{w} = \mathbf{y}$.
- Minimum-norm inductive bias ($\hat{\mathbf{w}} = \mathbf{X}^\dagger \mathbf{y}$) selects the minimum-energy interpolator.
- Overparameterization distributes energy across more dimensions: $\|\hat{\mathbf{w}}\|$ contracts, driving Var $\downarrow$.

The bias-variance decomposition $\sigma^2 + \text{bias}^2 + \text{Var}$ holds unconditionally; the classical U-curve is specific to underparameterization, unified by double descent in overparameterized models.

Recap: structural principles of the bias–variance framework

- **1. Exact Pythagorean decomposition:** Expected squared error splits unconditionally into irreducible noise, approximation bias, and estimation variance.

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- **3. Orthogonal error mechanisms:** bias^2 is an approximation property of hypothesis space $\mathcal{H}$; Var is a finite-sample property of training data S .
- **4. Variance control via regularization:** L_2 regularization trades minimal bias for massive variance suppression; increasing sample size N shrinks variance unilaterally.

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- 1 Validation
- 2 Bias–Variance
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Binary classification as statistical decision theory I

Let $X \in \mathcal{X}$ and binary response $Y \in \{0, 1\}$. The conditional expectation coincides with the posterior probability:

$$q(x) := \mathbb{E}[Y \mid X = x] = \Pr(Y = 1 \mid X = x).$$

Binary classification as statistical decision theory II

Decision Rule and Expected 0/1 Risk:

- A classifier is a rule $d : \mathcal{X} \rightarrow \{0, 1\}$. Under 0/1 loss $\ell_{0/1}(y, d(x)) = \mathbf{1}\{d(x) \neq y\}$:

$$R(d) = \mathbb{E}_{(X,Y)} [\mathbf{1}\{d(X) \neq Y\}] = \mathbb{E}_X \left[\Pr(d(X) \neq Y \mid X) \right].$$

- **Pointwise risk decomposition:** $\Pr(d(x) \neq Y \mid x) = \mathbf{1}\{d(x) = 1\}(1 - q(x)) + \mathbf{1}\{d(x) = 0\}q(x)$.

Binary classification reduces to statistical estimation of posterior probability $q(x)$, followed by pointwise risk-minimizing thresholding.

Lemma: The Bayes optimal decision rule and Bayes risk I

Bayes Optimal Classifier. Under symmetric 0/1 loss, the decision rule that minimizes expected risk pointwise is:

$$d^*(x) = \mathbf{1}\left\{q(x) \geq \frac{1}{2}\right\}.$$

Lemma: The Bayes optimal decision rule and Bayes risk II

Generalization to Asymmetric Cost Matrices:

- Assign costs: c_{10} for false positive ($d = 1, y = 0$), c_{01} for false negative ($d = 0, y = 1$).
- The conditional expected risks evaluate to:

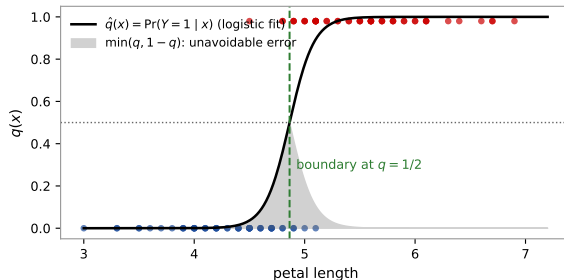
$$R(d = 1 \mid x) = c_{10}(1 - q(x)), \quad R(d = 0 \mid x) = c_{01}q(x).$$

- Setting $R(d = 1 \mid x) \leq R(d = 0 \mid x)$ yields the optimal decision rule analytically:

$$d^*(x) = \mathbf{1} \left\{ q(x) \geq \frac{c_{10}}{c_{01} + c_{10}} \right\}.$$

Statistical estimation of $q(x)$ is completely decoupled from operational cost matrices:
estimate probabilities first, then threshold according to mission-specific costs.

Living case: the Bayes error as irreducible class overlap



The shaded region represents irreducible Bayes risk R^* ;
it quantifies the intrinsic overlap between class-conditional distributions that no classifier can
eliminate.

Q: Why ordinary least squares fails for classification

Why not simply fit binary response $y \in \{0, 1\}$ using linear regression $\mathbf{w}^\top \mathbf{x}$?

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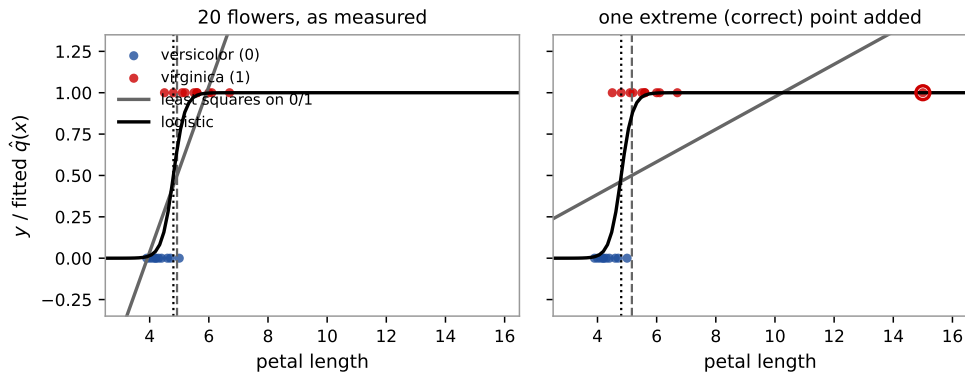
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Squared error penalizes margin surplus as an error;
proper classification requires a loss function that saturates on confident correct samples.

Living case: failure of least squares under margin leverage



Extreme positive margins distort the least squares boundary, doubling test errors;
in contrast, logistic regression is robust because confident correct samples generate near-zero gradients.

Lemma: The sigmoid as the canonical inverse link I

We require a smooth monotonic mapping from unbounded linear score $z = \mathbf{w}^\top \mathbf{x} \in \mathbb{R}$ to probability $\mu \in (0, 1)$.

Lemma: The sigmoid as the canonical inverse link II

Derivation of the Canonical Inverse Link:

- In Generalized Linear Models (GLMs), equating the linear score $z = \mathbf{w}^\top \mathbf{x}$ to the natural parameter η defines the **canonical link**:

$$\ln \frac{q(x)}{1 - q(x)} = \mathbf{w}^\top \mathbf{x}.$$

- Inverting the logit operator yields the logistic sigmoid:

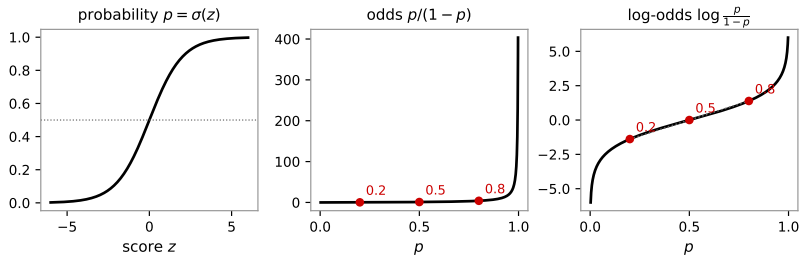
$$q(x) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}} =: \sigma(\mathbf{w}^\top \mathbf{x}).$$

- **Derivative property:** $\sigma'(z) = \sigma(z)(1 - \sigma(z)) = p(1 - p).$

Lemma: The sigmoid as the canonical inverse link III

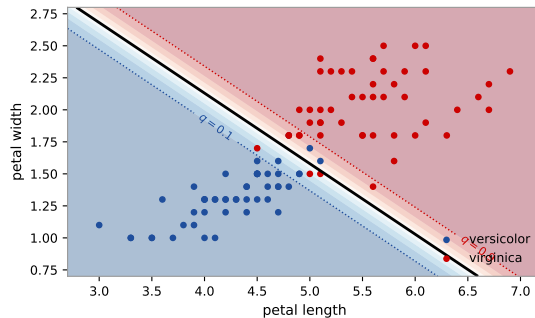
The sigmoid is not an arbitrary heuristic S-curve;
it is the unique **canonical inverse link function** of the Bernoulli exponential family.

Living case: the log-odds coordinate system



The log-odds transform maps bounded probabilities $(0, 1)$ homeomorphically to $\mathbb{R}$, converting multiplicative likelihood ratios into additive feature contributions: $\mathbf{w}^\top \mathbf{x} = \sum_j w_j x_j$.

Living case: decision boundary and posterior uncertainty



A linear hyperplane $\mathbf{w}^\top \mathbf{x} = 0$ defines the decision boundary $\hat{q} = 0.5$, while the sigmoid function induces calibrated spatial contours of posterior classification uncertainty.

Recap: theoretical foundations of probabilistic classification

- **Target equivalence:** Binary classification estimates the identical population conditional expectation $q(x) = \mathbb{E}[Y \mid X = x]$ as regression.

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- **Exponential family derivation:** Modeling the natural parameter linearly uniquely determines the logistic sigmoid $\sigma(\mathbf{w}^\top \mathbf{x})$ via the canonical link of the Bernoulli family.
- **Saturation and robustness:** Probabilistic modeling replaces quadratic margin distortion with bounded residual gradients, ensuring robustness to extreme margin outliers.

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Maximum likelihood estimation as information projection I

Given empirical sample $S = \{(x_n, y_n)\}_{n=1}^N \sim \mathcal{D}^N$, conditional Bernoulli likelihood is:

$$L_N(\mathbf{w}) := \prod_{n=1}^N p_{\mathbf{w}}(y_n \mid x_n) = \prod_{n=1}^N (p_n)^{y_n} (1 - p_n)^{1-y_n}.$$

Maximum likelihood estimation as information projection II

Decomposition via Kullback–Leibler Divergence:

$$\begin{aligned}\hat{R}_N(\mathbf{w}) &= \mathbb{E}_{\hat{p}} \left[-\ln p_{\mathbf{w}}(Y | X) \right] \\ &= \underbrace{D_{\text{KL}}(\hat{p}(y | x) \parallel p_{\mathbf{w}}(y | x))}_{\text{divergence to parametric hypothesis}} + \underbrace{H(\hat{p})}_{\text{empirical data entropy}}.\end{aligned}$$

The empirical entropy $H(\hat{p}) = -\frac{1}{N} \sum_n \ln \hat{p}(y_n | x_n)$ is invariant to parameter vector $\mathbf{w}$:

$$\arg \min_{\mathbf{w}} \hat{R}_N(\mathbf{w}) \equiv \arg \min_{\mathbf{w}} D_{\text{KL}}(\hat{p} \parallel p_{\mathbf{w}}).$$

Minimizing cross-entropy is the exact **information projection** of empirical data $\hat{p}$ onto the parametric model manifold under KL divergence.

Lemma: Closed-form loss in the logit score representation I

Parameterize the per-point loss directly in the linear score $z_n := \mathbf{w}^\top \mathbf{x}_n$, with predicted probability $p_n = \sigma(z_n)$:

$$\begin{aligned}\ell(z_n, y_n) &= -y_n \ln \sigma(z_n) - (1 - y_n) \ln (1 - \sigma(z_n)) \\ &= -y_n \ln \frac{e^{z_n}}{1 + e^{z_n}} - (1 - y_n) \ln \frac{1}{1 + e^{z_n}} \\ &= \ln(1 + e^{z_n}) - y_n z_n.\end{aligned}$$

The term $\text{softplus}(z) := \ln(1 + e^z)$ is the smooth analytic approximation to $\max(0, z)$.

Lemma: Closed-form loss in the logit score representation II

Two Fundamental Computational Advantages:

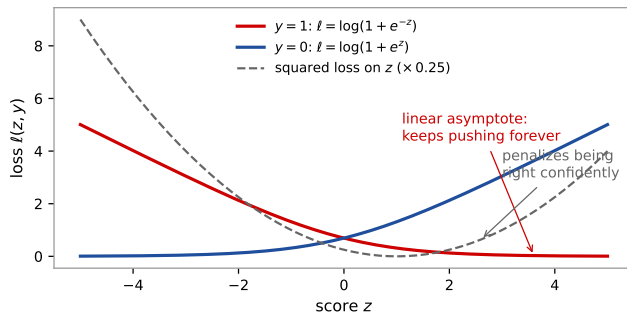
1. Floating-Point Stability. Evaluating loss via saturated probabilities $p_n \in \{0, 1\}$ triggers numerical underflow ($\ln 0 = -\infty$). The softplus form $\ln(1 + e^z) - yz$ evaluates stably across all real scores $z \in \mathbb{R}$.

2. Functional Margin Formulation. For bipolar labels $y_n \in \{-1, +1\}$ with functional margin $m_n := y_n z_n$:

$$\ell(m_n) = \ln(1 + e^{-m_n}).$$

Formulating cross-entropy as $\ell(z, y) = \ln(1 + e^z) - yz$ eliminates floating-point instability and establishes global convexity.

Living case: margin asymptotics of the logistic loss



Logistic loss decays smoothly to zero for positive margins ($z \rightarrow \infty$) without margin distortion, while maintaining a linear slope for errors ($z \rightarrow -\infty$) to sustain optimization gradients.

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Q1: How Trustworthy is a Validation Score? I

A computed validation score $L_{S_{\text{val}}}(f)$ is a single finite-sample scalar. How faithfully does it reflect the true population risk $L_{\mathcal{D}}(f)$?

1. Non-Asymptotic Concentration (Pointwise Guarantee):

Hoeffding Sub-Gaussian Interval. For a candidate f fixed *prior* to observing S_{val} , and bounded loss $\ell \in [a, b]$, with probability at least $1 - \delta$:

$$|L_{S_{\text{val}}}(f) - L_{\mathcal{D}}(f)| \leq (b - a) \sqrt{\frac{\ln(2/\delta)}{2m}}.$$

A validation score is an **interval estimate**, not a point: confidence dependence is mild ($\sqrt{\ln(2/\delta)}$), but narrowing the margin requires quadratic sample complexity ($m \propto 1/\epsilon^2$).

Q1: How Trustworthy is a Validation Score? II

2. Structural Optimism under Model Selection (Winner Coupling):

Q2: What Really Constitutes Generalization Error? I

When retraining on independent training sets $S \sim \mathcal{D}^N$, predictions fluctuate. What fundamental statistical mechanisms govern the expected test error?

1. Pythagorean Orthogonality in L_2 Hilbert Space:

Three-Term Orthogonal Identity. Centered at the ensemble mean $\bar{f}(x_0) := \mathbb{E}_S[f_S(x_0)]$, squared loss decomposes unconditionally:

$$\mathbb{E}_{S, \varepsilon} \left[(y_0 - f_S(x_0))^2 \right] = \underbrace{\sigma^2}_{\text{irreducible noise}} + \underbrace{(f(x_0) - \bar{f}(x_0))^2}_{\text{bias}^2(x_0) \text{ (approximation error)}} + \underbrace{\mathbb{E}_S \left[(f_S(x_0) - \bar{f}(x_0))^2 \right]}_{\text{Var}(x_0) \text{ (finite-sample variance)}} .$$

Q2: What Really Constitutes Generalization Error? II

Cross-terms vanish identically because query noise, deterministic approximation bias, and zero-mean sample fluctuations inhabit mutually orthogonal subspaces.

Q2: What Really Constitutes Generalization Error? III

2. Physical Origins and the Modern Double Descent Perspective:

Q2: What Really Constitutes Generalization Error? IV

Generalization error reflects the synthesis of **aleatoric noise, hypothesis bias, and sample variance**;
regularization controls this balance by trading minor bias for orders-of-magnitude variance reduction.

Q3: Why Estimate Probabilities Before Applying the Sigmoid? I

Classification assigns discrete labels $y \in \{0, 1\}$. Why detour through continuous posterior probabilities $q(x)$, and why does the Sigmoid appear universally?

1. Target Equivalence and Decoupled Decision Theory: Regression and classification share the exact same statistical estimand—the conditional expectation:

$$q(x) := \mathbb{E}[Y \mid X = x] = \Pr(Y = 1 \mid X = x).$$

Estimating $q(x)$ is purely statistical; discrete assignment $d(x) \in \{0, 1\}$ belongs to decision theory.

Q3: Why Estimate Probabilities Before Applying the Sigmoid? II

Bayes Cost-Sensitive Decision Rule. Under asymmetric operational costs (c_{10} for false positive, c_{01} for false negative), the optimal action thresholds posterior probability analytically:

$$d^*(x) = \mathbf{1} \left\{ q(x) \geq \frac{c_{10}}{c_{01} + c_{10}} \right\}.$$

Estimating $q(x)$ decouples learning from operational utility: shifting costs requires only adjusting the threshold without retraining the model.

Q3: Why Estimate Probabilities Before Applying the Sigmoid? III

2. The Sigmoid as the Canonical Inverse Link of Bernoulli GLMs: The Sigmoid is not an ad-hoc heuristic S-curve; it is an algebraic necessity of exponential families. The Bernoulli distribution expressed in canonical exponential form:

$$p(y \mid \mu) = \exp \left(y \ln \frac{\mu}{1 - \mu} + \ln(1 - \mu) \right) \implies \text{natural parameter } \eta := \ln \frac{\mu}{1 - \mu} = \text{logit}(\mu).$$

Generalized Linear Models equate natural parameter to linear score $\eta := \mathbf{w}^\top \mathbf{x}$. Inverting the canonical link yields the **Logistic Sigmoid**:

$$q(x) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}.$$

Probability estimation cleanly decouples statistical inference from operational cost matrices; the Sigmoid is the unique canonical inverse link function linearizing Bernoulli natural parameters.

Q4: Why is Cross-Entropy the “Correct” Classification Loss? I

Why avoid least squares $(y - \sigma(\mathbf{w}^\top \mathbf{x}))^2$ on binary targets? Why has cross-entropy $\ln(1 + e^z) - yz$ become the foundational standard for probabilistic classification?

1. Information Projection via Kullback–Leibler Divergence: Maximizing Bernoulli likelihood is mathematically identical to minimizing KL divergence from empirical distribution $\hat{p}$ to parametric model $p_{\mathbf{w}}$:

$$\hat{R}_N(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (-\ln p_{\mathbf{w}}(y_n | x_n)) = D_{\text{KL}}(\hat{p} \parallel p_{\mathbf{w}}) + H(\hat{p}).$$

Cross-entropy is a strictly proper scoring rule: its population minimizer uniquely recovers true posterior $q(x)$.

Q4: Why is Cross-Entropy the “Correct” Classification Loss? II

2. Self-Regulating Residual Gradients:

Q4: Why is Cross-Entropy the “Correct” Classification Loss? III

3. Global Convexity, Fisher Curvature, and L_2 Coercivity: The Hessian $\mathbf{H}(\mathbf{w}) = \frac{1}{N} \mathbf{X}^\top \mathbf{D} \mathbf{X} \geq \mathbf{0}$ ensures absence of spurious local minima, with curvature matching empirical Fisher information. On linearly separable data where finite MLE does not exist ($\|\mathbf{w}\| \rightarrow \infty$), Tikhonov L_2 regularization induces coercivity and uniform positive curvature:

$$\mathbf{H}_\lambda(\mathbf{w}) = \frac{1}{N} \mathbf{X}^\top \mathbf{D}(\mathbf{w}) \mathbf{X} + \lambda \mathbf{I} \geq \lambda \mathbf{I} > \mathbf{0},$$

guaranteeing existence and uniqueness of the global minimizer by the Weierstrass Extreme Value Theorem.

Cross-entropy represents canonical information projection, yielding residual gradients and Fisher curvature;

paired with L_2 regularization, it establishes a mathematically complete framework for classification.